# ME24101 BME Module 5

**Renewable** energy is energy obtained from naturally repetitive and persistent flows of energy occurring in the local environment. The renewable energy systems covered include power from solar radiation (sunshine), wind, biomass (plant crops), river (hydropower), Ocean waves, tides, geothermal heat, and other such continuing resources.

Renewable sources

Renewable technologies are presently under development stage. At present its share is very small.

i. Solar energy- Solar energy can be major source of power and can be harnessed by using thermal and photovoltaic conversion systems. The earth continuously intercepts solar power of 178 billion MW, which is about 10,000 times the world's demand. According to one estimate, if all the buildings of the world are covered with solar panel, it can fulfill electrical power requirements of the world.
Solar PV power is considered as an expensive source of power. The use of solar energy is growing strongly around the world due to the rapidly declining solar panel manufacturing costs.

ii. Wind energy- The power available in the winds over the earth surface is estimated to be $1.6 \times 10^7$ MW, which is more than the present energy requirement of the world. Wind power has emerged as the most economical of all renewable energy sources. The installation cost of wind power is Rs. 4 crore / MW, which is comparable to that of conventional thermal plants.

iii. Hydro resources- Among all renewables, hydropower is the most advanced and flexible source of power. It is well developed and established source of electric power. Due to requirement of huge capital investment and strong environmental concerns about large plants, only about one third of the realistic potential has been taped so far. Hydro installation and plants are long lasting due to continuous steady operation without high temperature and stress. Therefore, it often produces electricity at low cost with consequent economic benefits.

iv. Biomass energy- Energy resources available from animal and vegetation are called biomass energy resources. Historically, human have harnessed biomass-derived energy since the time when people began burning wood to make fire. Even today, biomass is the only source of fuel for domestic use in many developing countries. The principal biomass resources are trees (wood, leaves and forest industry waste), cultivated plants grown for energy, algae and other vegetation from ocean and lake, urban waste (municipal and industrial waste) and rural waste (agricultural and animal waste, crop, residue etc). Solar energy absorbed by plants (through photosynthesis process) is estimated to be $2 \times 10^{21}$ J / year. Biomass material may be transformed by chemical or biological processes to other usable intermediate bio-fuels such as biogas (methane), ethanol, biodiesel and charcoal etc.

v. Geothermal energy- Geothermal energy comes from the natural heat of the Earth primarily due to the decay of the naturally radioactive isotopes of Uranium, Thorium and Potassium. Geothermal energy is derived from huge amount of stored thermal energy in the interior of earth. However, its economic recovery on the surface of the earth is not feasible everywhere. Its overall contribution in total energy requirement is negligible. However, it is a very important resource locally. Almost all the countries use it directly but 24 countries use it for electricity generation.

vi. Ocean tidal energy- Tidal energy is a form of hydropower that converts energy of tides into electricity or other useful forms of power. Although not yet widely used, tidal power has potential for future electricity generation. Tides are more predictable than wind energy and solar power. It is in developing stage. There are at present only few operational tidal power plants.

vii. Ocean wave energy- Wave power refers to the energy of ocean surface waves and the capture of that energy to do useful work. Good wave power locations have a flux of about 50 kW / m of shoreline. As per an estimate, the potential for shoreline-based wave power is about 50,000 MW. Deep-water wave power resources are truly enormous, but perhaps impractical to capture. The worldwide resource of wave energy has been estimated to be greater than 2 TW.

viii. Ocean thermal energy conversion- OTEC technology is in infant stage. Conceptual design of small OTEC plant has been finalized. Their commercial prospects are quite uncertain. The potential is likely to be more than that of tidal or wave energy. The resource potential for OTEC is considered to be much larger than for other ocean energy forms. Up to 88,000 TWh /yr of power could be generated from OTEC without affecting the ocean's thermal structure. While OTEC technology has been demonstrated in the ocean for many years, the engineering and commercial challenges have constrained the development of the technology.

Benefits:

i. Renewable energy emits no or low greenhouse gases. That's good for the climate.
The combustion of fossil fuels for energy results in a significant amount of greenhouse gas emissions responsible for global warming. Most sources of renewable energy result in little to no emissions, even when considering the full life cycle of the technologies.

ii. Renewable energy emits no or low air pollutants. That's better for our health.
A worldwide increase in fossil fuel-based road transport, industrial activity, and power generation (as well as the open burning of waste in many cities) contributes to

elevated levels of air pollution. In many developing countries, the use of charcoal and fuel-wood for heating and cooking also contributes to poor indoor air quality. Particles and other air pollutants from fossil fuels literally asphyxiate cities. According to studies by the World Health Organization, their presence above urban skies is responsible for millions of premature deaths and costs billions.
Instead of depleting precious resources and polluting the environment, renewable energy meets the objectives of a circular economy and is a strong motor for social and economic development.

iii. Renewable energy comes with low costs. That's good for keeping energy prices at affordable levels.
Geopolitical strife and upheavals often come with increasing energy prices and limited access to resources. Since renewable energy is produced locally, it is less affected by geopolitical crisis or price spikes or sudden disruptions in the supply chain.

iv. Renewable energy creates jobs. That's good for the local community.
Renewables make urban energy infrastructures more independent from remote sources and grids. Businesses and industry invest in renewable energy to avoid disruptions, including resilience to weather-related impacts of climate change.

v. Renewable energy is accessible to all. That's good for development.
In many parts of the world, renewables represent the lowest-cost source of new power generation technology, and costs continue to decline. Especially for cities in middle- and low-income countries, renewable energy is the only way to expand energy access to all inhabitants, particularly those living in urban slums and informal settlements, in suburban and peri-urban areas and in remote areas.

vi. Renewable energy is secure. That's good for stability.
Evolving energy markets and geopolitical uncertainty have moved energy security and energy infrastructure resilience to the forefront of many national energy strategies. Security of supply is a serious concern in energy markets worldwide, from the European Union and the United States to Egypt and India.

vii. Renewable energy is democratic. That's good for acceptance.
In recent years, the number of community energy projects using renewable sources, have surged in various parts of the world. Although community energy is frequently associated with Northern European countries such as Denmark and Germany, such projects are emerging in other parts of the world including Thailand, Japan, and Canada. This trend confirms that democracy is an important driver for the change to renewable energy.

Limitation:

i. Weather Dependency – Non-conventional sources of energy such as solar and wind power rely heavily on the weather, which can make it difficult to generate consistent and reliable power.

ii. High Initial Investment – The initial investment required to set up non-conventional energy systems can be quite high, which may not be feasible for some individuals or organizations.

iii. Inefficient Storage – Non-conventional energy sources such as solar and wind power are often generated in large amounts during peak times, but storage options for this excess energy may not be efficient or cost-effective.

iv. Limited Capacity – Non-conventional sources of energy have a limited capacity when compared to conventional sources such as coal and natural gas, which can limit their overall potential.

v. Environmental Impact – Non-conventional energy sources can also have negative environmental impacts, such as large-scale hydroelectric dams that can cause habitat destruction and displace local communities.

**Non-renewable** energy is energy obtained from static stores of energy that remain underground unless released by human interaction. Examples are nuclear fuels and the fossil fuels of coal, oil and natural gas. With these sources, the energy is initially an isolated energy potential and external action is required to initiate the supply of energy for practical purposes. Such energy supplies are called finite supplies or brown energy.

Non-renewable sources

i. Fossil fuel- India has vast reserves of coal, 5$^{th}$ largest in the world after the USA, Russia, China and Australia. According to rough estimate, total recoverable coal in India is 60,000 million tons, about 6.8% of world's total. Our coal production is not enough to meet the growing consumption. Net coal import dependency has risen from negligible percentage in 1990 to nearly 23% in 2014.We have very little oil and gas reserves. The estimated reserve of crude oil is 762 MT, which is 0.34% of global oil reserves. Likewise the estimated reserve of Natural gas is 1427 bcm, which is 0.68% of global gas reserves. Oil and gas reserves are insufficient even for transportation sector. India at present imports 75% of oil and 25% of natural gas. Largest consumer of natural gas is fertilizer industry followed by power industry.

ii. Nuclear resources- India has modest (1 to 2% of global) reserves of Uranium, mostly located at Jadugoda, Jharkhand. There are 21 nuclear reactors at 7 plants located at Narora, Rawatbhatta, Kakrapar, Kaiga, Kalpakkam, Kudanulam and Trombay (18 PHWR, 2 BWR and 1 VVER). Present, nuclear power installed capacity is 5780 MW, which is 2.1% of total installed electricity generating capacity. The government has established a research facility IGCAR (India Gandhi Center for Atomic Research) way back in 1971 at Kalpakkam, Tamil Nadu for development of the nuclear energy technology.

Benefits:

i. Cost- At present these are cheaper than renewable sources.
ii. Efficiency- The efficiency of the energy sources is high. Because from 1gm of uranium we get 1 MW energy, from 1-tonne coal we get 2460 kWh energy.
iii. Security- As storage is easy and convenient, by storing certain quantity, the energy availability can be ensured for certain period.
iv. Convenience- These sources are very convenient to use as technology for their conversion and use is universally available.
v. Transport- The raw materials of conventional Sources of Energy are easy to transport. Raw materials such as coal, petroleum, natural gas can be transported easily through trains or ships from one place to another.
vi. Location- Generally it doesn't need any specific place for installation. The government can easily set up a conventional plant according to their requirements. Such as if the government wants to install a thermal plant in Uttarakhand or in Jammu they can easily install it.

Limitation:

i. Pollution- They are the main reason for the pollution because, it releases carbon monoxide from polluters into the atmosphere. According to The International Energy Agency, only in 2018, India emitted 2,299 million tonnes of carbon monoxide. This report also said that India's per capita emissions were about 40% of the global average and contributed 7% to the global carbon dioxide burden.
Nuclear plants generate radioactive waste. We all know about the Fukushima Daiichi nuclear disaster which happened in 2011. Recently Japanese government has decided to release radioactively contaminated water into the ocean. But the environmentalists warn that it should be harmful to ocean life.

ii. Lifetime of supply – Sources of energy are non-renewable will eventually run out. Once the resources are depleted, they cannot be replaced.

iii. High startup cost- According to the government estimate, the cost may be about 50 to ₹70 crores for setting up a 10 MW thermal power unit, whereas to set up a nuclear power plant, it is required ₹60,000 crores.

<u>Comparison of renewable and conventional energy systems</u>

| Sl No. | | Renewable energy supplies (green) | Conventional energy supplies (brown) |
|---|---|---|---|
| 1 | Examples | Wind, solar, biomass, tidal | Coal, oil, gas, radio-active ore |
| 2 | Source | Natural local environment | Concentrated stock |
| 3 | Normal state | A current or flow of energy. An income | Static store of energy. Capital |

| | | | |
|---|---|---|---|
| 4 | Lifetime of supply | Infinite | Finite |
| 5 | Cost of source | Free | Increasingly expensive |
| 6 | Equipment capital cost per kW capacity | Expensive, commonly ≈ $1000 | Moderate, perhaps $500 without emissions control; yet > $1000 with emissions reduction |
| 7 | Location for use | Site and society specific | General and global use |
| 8 | Scale | Small-scale is often economic | Increased scale often improves supply costs; large-scale frequently favored |
| 9 | Skills | Interdisciplinary and varied. Wide range of skills. Importance of bioscience and agriculture | Strong links with electrical and mechanical engineering. Narrow range of personal skills |
| 10 | Context | Well adapted to rural situations and decentralized industry | Scale favors urban, centralized industry |
| 11 | Safety | Local hazards possible in operation. Usually safe when out of action | May be shielded and enclosed to lessen great potential dangers; most dangerous when faulty |
| 12 | Pollution and environmental damage | Usually little environmental harm, especially at moderate scale.<br>Hazards from excessive wood burning. Soil erosion from excessive bio-fuel use. Large hydro reservoirs disruptive. | Environmental pollution common, especially of air and water.<br>Permanent damage common from mining and radio-active elements entering water table.<br>Deforestation and ecological sterilization from excessive air pollution.<br>Greenhouse gas emissions causing climate change |

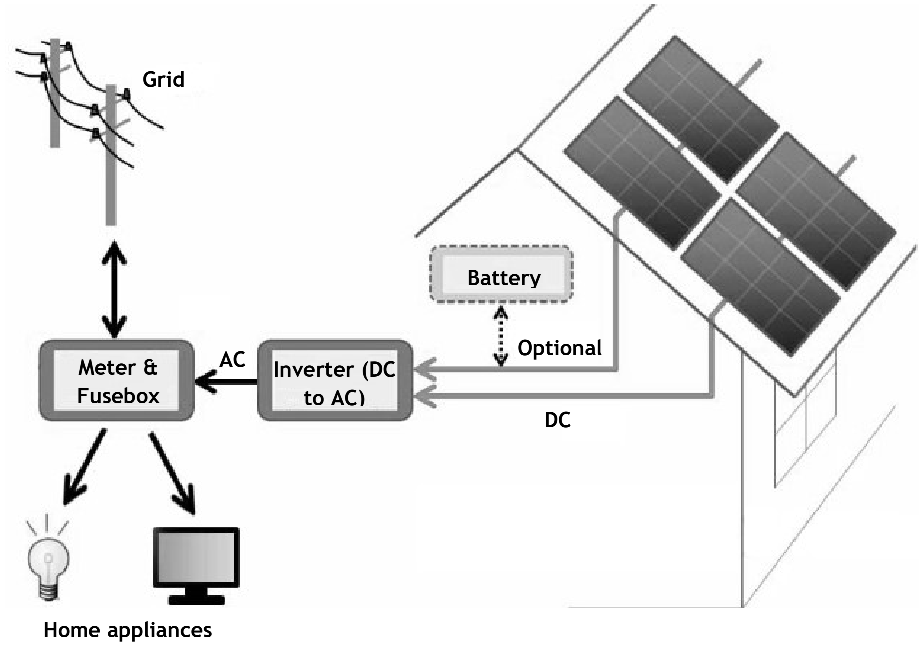
**Solar energy**

Solar energy is the radiant light and heat from the sun that has been harnessed by humans since ancient times using a range of ever-evolving technologies. Solar radiation accounts for most of the available renewable energy on earth. All other renewable energies other than geothermal derive their energy from energy received from the sun.

<u>Solar-Energy Power generation system</u>

Electrical energy can be harvested from sunlight (solar radiation) by means of either photovoltaic or concentrated solar power system.

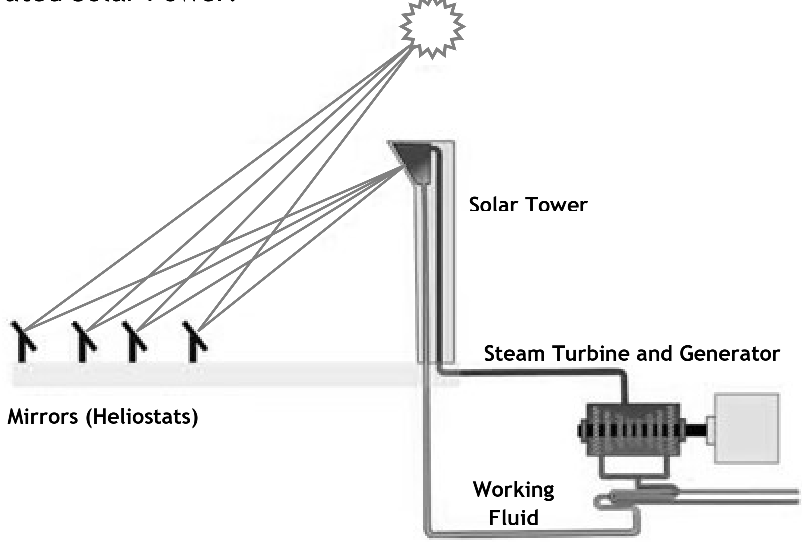
Solar Photovoltaic (PV):




Photovoltaic directly converts solar energy into electricity. It works on the principle of the photovoltaic effect. Solar cells (PV cells) are made of p-type and n-type silicon semiconductors. When the cells are exposed to sunlight, they absorb photons and release free electrons. The bus bars collect this running electron which results in electric current. This phenomenon is called as the photoelectric effect. Photovoltaic effect is a method of producing direct current electricity based on the principle of the photoelectric effect.

Based on the principle of photovoltaic effect, solar cells or photovoltaic cells are made. They convert sunlight into direct current (DC) electricity. But, a single photovoltaic cell does not produce enough amount of electricity. Therefore, a number of photovoltaic cells are mounted on a supporting frame and are electrically connected to each other to form a photovoltaic module or solar panel. As of now it is clear that photovoltaic modules produce DC electricity. But, for most of the times we require AC power and, hence, solar power system consists of an inverter too.

Concentrated Solar Power:



As the name suggest, in this type of solar power system, sun rays are concentrated (focused) on a small area by placing mirrors or lenses over a large area. Due to this, a huge amount of heat is generated at the focused area. This heat can be used to heat up the working fluid which can further drive the steam turbine. There are different types of technologies that are based on the concentrated solar power to produce electricity. Some of them are - parabolic trough, Stirling dish, solar power tower etc. The following schematic shows how a solar power tower works.

Benefits of Solar Power:

i. Solar power is a renewable energy source. It is available everyday and can be harnessed in different parts of the world. Unlike other sources of energy, solar energy will not run out. As long as there is the sun, solar energy is accessible for us.

ii. Solar energy does not emit hazardous gas like carbon. It also reduces the greenhouse effect. It avoids the environmental damage associated with mining or drilling for fossil fuels. Furthermore, solar energy also uses little to no water, unlike power plants that generate electricity using steam turbines.

iii. If we get solar panels, then it means that we have a chance to create our own source of energy. So, we can take advantage of energy independence. We do not need to be dependent or rely on the traditional electric grid. We have control over our energy source. The excess power can be sold to the utility.

iv. Solar systems are fairly easy to install and require very little maintenance. Solar panels have no moving parts that wear out over time. Just keep them clean and in good physical condition to keep them working properly. Between their low maintenance costs and average lifespan of 25 years, it can be easy to get our money's worth when investing in solar panels.

v. Solar energy systems can generate electricity in any climate. Cloudy days reduce the amount of electricity. Cold, however, doesn't affect productivity. Snowfall can actually help solar system, as the snow cleans the panels when it melts and sun reflected off the snow increases the amount of light hitting panels. The result is more electricity production.

Disadvantages:

i. One of the disadvantages of solar power is that it comes with a high initial cost of purchase and installation. This includes wiring, batteries, inverter, and solar panels.

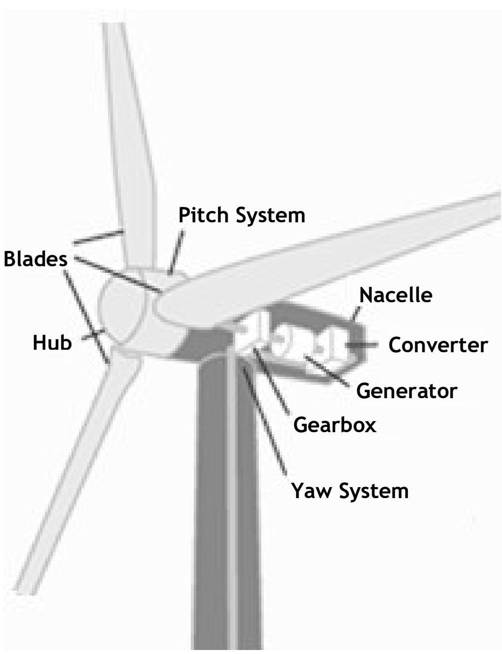
ii. Being an intermittent energy source is another drawback of solar energy. At night, the sun does not shine. So, the solar panels cannot generate power. Depending on our location, time of the day, and even time of the year, the sun's intensity varies.

iii. In order to produce more electricity, more solar panels are demanded to collect more sunlight. Compared to fossil fuels, solar panels have low power density. So, a larger area for solar panels is needed for similar amounts of energy production.

iv. While solar power system pollution related issues are less compared to other energy sources, there is still a bit of pollution to be aware of. There are also some toxic and hazardous products used during the manufacturing process of solar photovoltaic systems, which can indirectly affect the environment. Transportation and installation of solar systems are associated with the emission of greenhouse gases. The pollution and emissions are saved after they are installed.

v. Depending on geographical location the size of the solar panels vary for the same power generation.

**Wind energy**




Wind turbines convert the kinetic energy in the wind into mechanical power. A generator can convert mechanical power into electricity. The mechanical power can also be utilized directly for specific tasks such as pumping water.
There are two basic types of wind turbines: those with a horizontal axis, and those with a vertical axis. The majority of wind turbines have a horizontal axis.

**Basic components of a wind turbine include**:

i. A rotor- consists of the blades and the hub which convert the wind energy into rotational shaft energy.

ii. A nacelle containing a drive train- includes shafts, gearbox and generator.

iii. Pitch drive- turns the blades out of the wind to control rotor speed. Turbines catch the wind's energy with their propeller-like blades, which act much like an airplane wing. The turbine blades are swiveled depending upon the speed of the wind. The technique is called pitch control. It provides the best possible orientation of the turbine blades along the direction of the wind to obtain optimized wind power.

iv. Yaw drive- keeps the rotor and therefore the turbines facing the wind. The orientation of the nacelle or the entire body of the turbine can follow the direction of changing wind direction to maximize mechanical energy harvesting from the wind. The direction of the wind along with its speed is sensed by an anemometer (automatic speed measuring devices) with wind vanes attached to the back top of the nacelle. The signal is fed back to an electronic microprocessor-based controlling system which governs the yaw motor which rotates the entire nacelle with gearing arrangement to face the air turbine along the direction of the wind.

v. Brake- is housed in the nacelle to stop or slow the rotor mechanically, electrically or hydraulically in emergencies.

vi. A tower- supports the rotor and drive train; electronic equipment such as controls, electrical cables, ground support equipment, and interconnection equipment

Working Principle:

When the wind blows, a pocket of low-pressure air is formed on one side of the blade. The low-pressure air pocket then pulls the blade toward it, causing the rotor to turn. This is called lift. The force of the lift is much stronger than the wind's force against the front side of the blade, which is called drag. The combination of lift and drag causes the rotor to spin like a propeller.

The rotor shaft, known as low speed shaft connects gearbox. In gearbox, a series of gears increase the rotation of the rotor from about 18 revolutions a minute to roughly 1,800 revolutions per minute. The high speed shaft from the gearbox is coupled with the rotor of the generator and hence the electrical generator runs at a higher speed and produces necessary electricity.

Benefits:

i. Wind is almost everywhere. Since it is well known where the wind blows strongest, it is possible to install wind plants based on our considerable awareness of the local conditions.

ii. Wind power is excellent in remote areas. In more isolated areas far from towns or cities, where there is no connection to the electricity grid, it can be exploited using micro-grid solutions. This offers an important opportunity because it can lead to significant savings from not having to build expensive infrastructure.

iii. The transformation of wind power into electricity has impressive. Efficiency ranges from 40% to 50%. This conversion efficiency is very close to the maximum theoretical level.

iv. Wind is a type of clean energy. It does not emit greenhouse gasses when generating electricity. It can minimize the use of burning fossil fuels and thus can reduce carbon footprint.

v. Wind is a renewable energy source. It comes from wind, which is a naturally occurring resource that does not get used up.

vi. Wind power has a low operating cost. Wind power is a renewable energy source and hence there is no ongoing expense to acquire fuel. Once the wind turbine is installed, the only real cost is maintenance and it is minimal and inexpensive.

vii. Wind turbines save space. Because they are elevated off the ground, the space below them is open to other uses, like farming.

Disadvantages:

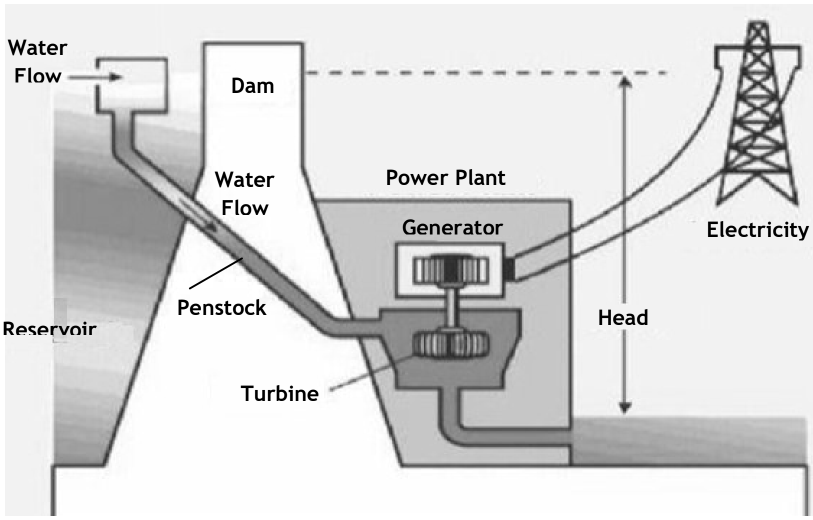
i. Wind turbines are known to pose a threat to the wildlife. Flying birds and bats, whose habitats or migratory paths could be injured or killed, if they run into the blades that turn on the fanlike structure of wind turbines when they are spinning. Aside from the wildlife that flies through the air, wildlife on the ground may also affected by the noise pollutions generated from whirring blades.

ii. In order to provide sufficient amount of electricity, a lot of land space is needed to construct wind turbines.

iii. Wind flow can be very unpredictable and without consistent wind flow, a wind turbine generator is useless. Even with ideal locations in coastal areas, hills, and open fields where the wind is especially strong, it doesn't blow all the time. Energy generation slows or stops when the wind slows or stops.

iv. Wind turbines are very susceptible to damage from lightning because of their tall and metallic form, which, in very few cases can be dangerous for nearby animals or people.

**Hydro-Energy**




Hydropower, or hydroelectric power, is a renewable source of energy that generates power by using a dam or diversion structure to alter the natural flow of a river or other body of water. Hydropower relies on the endless, constantly recharging system of the water cycle to produce electricity, using water as a fuel that is not reduced or eliminated in the process.

Working Principle:

In order to produce hydroelectricity, high rise dams are constructed on the river to obstruct the flow of water and thereby collect water in very large reservoirs. The water from high level in the dam moves downstream through penstock or pipes and falls to the blade in a turbine at the bottom of the dam. The turbine starts to spin, which, in turn, spins a generator that ultimately produces electricity, which is then fed into the electrical grid to power homes, businesses, and industries. Water when stored at a height in the dam has a large amount of potential energy which gets converted into the kinetic energy of flowing water when it is allowed to fall on the turbines it gets converted into mechanical energy of the turbine and finally gets converted into the electric energy by the generator.

Benefits:

i. Hydropower is completely renewable, which means it will never run out unless the water stops flowing.

ii. The creation of hydroelectricity does not release emissions into the atmosphere. This is, of course, the biggest appeal of any renewable energy source.

iii. Hydropower is, by far, the most reliable renewable energy available in the world. Unlike when the sun goes down or when the wind dies down, water usually has a constant and steady flow 24/7.

iv. Since hydropower is so reliable, hydro plants can actually adjust the flow of water. This allows the plant to produce more energy when it is required or reduce the energy output when it is not needed. This is something that no other renewable energy source can do.

v. In addition to being a clean and cost-effective form of energy, hydropower plants can provide backup power to the grid immediately during major electricity outages or disruptions.

vi. Hydropower also produces a number of benefits outside of electricity generation, such as flood control, irrigation support, and water supply.

Disadvantages:

i. Most hydropower plants are large projects that involve the construction of a reservoir, power-generating turbines, and building a dam. These require significant monetary investments.

ii. The dam cannot be worked on in any location. Therefore, the location is limited to the construction of hydro-plant. Also the location where the hydropower plant is being built should be financially and environmentally sustainable.

iii. Hydropower is a renewable source of energy but there are usually environmental impacts associated with building the power plants. To create a hydro plant, a running water source must be dammed. This prevents fish from reaching their breeding ground, which in turn affects any animal that relies on those fish for food. As the water stops flowing, riverside habitats begin to disappear. This can even remove animals from accessing water. Plants that are at the bottom of a reservoir begin to decompose when they die and release large quantities of carbon and methane.

iv. Hydropower is dependent on the amount of water in any given location. Thus, the performance of a hydro plant could be significantly affected by a drought.

v. When dams are built at higher elevations, they pose a serious risk to any town nearby that is below it. Due to excess rainfall, the dam can be collapsed and may harm the human life.

vi. The amount of water available to produce hydroelectric will vary depending on the weather conditions, and therefore electric production is unreliable.